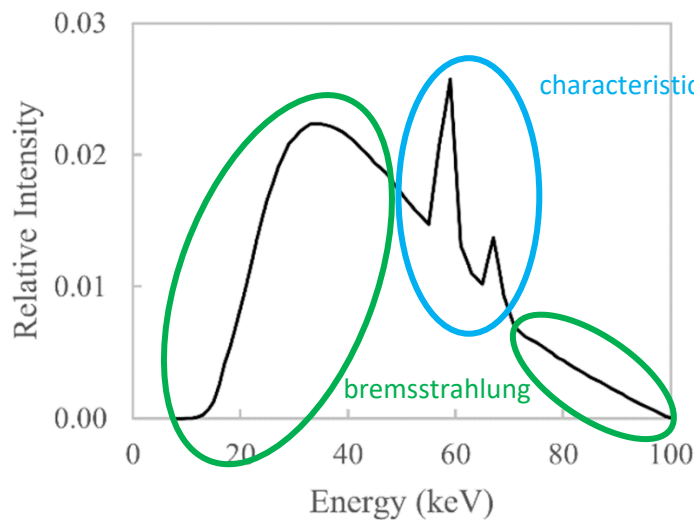


X-Ray, CT, and PET/SPECT

Q1: X-Ray (5 Points)

Below is a typical X-ray spectrum produced by an X-ray tube. The spectrum shows the intensity of X-rays (y-axis) as a function of their energy (x-axis).

- Identify and label the regions corresponding to bremsstrahlung radiation (1 Point)
- Identify and label the peaks corresponding to characteristic radiation (1 Point)



- Provide a brief explanation of how each type of radiation is produced and why they appear differently in the spectrum (3 Points):

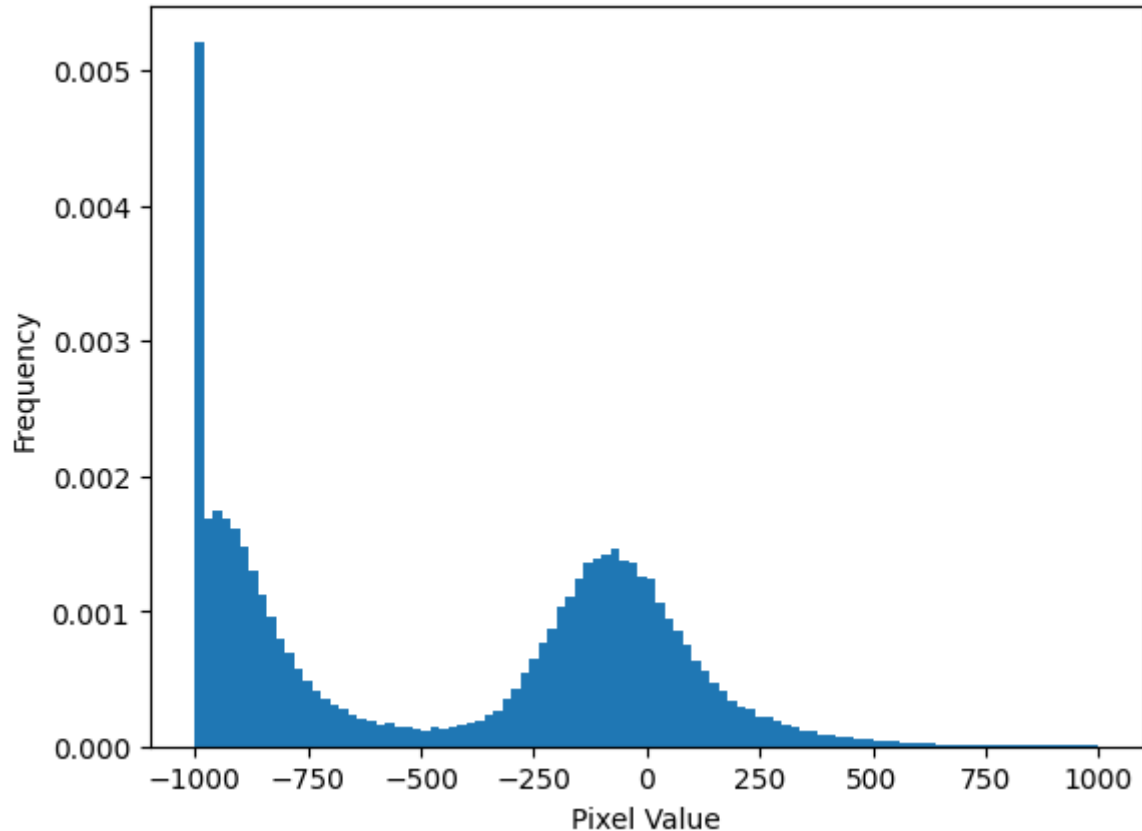
Characteristic radiation is produced when an incoming electron knocks out an inner-shell electron of a target atom, and an outer-shell electron fills the vacancy, releasing a photon with a specific energy unique to that atom—hence "characteristic."

Bremsstrahlung occurs when high-speed electrons are decelerated or deflected by the electric field of atomic nuclei, emitting a broad range of photon energies.

Characteristic radiation appears as sharp peaks in the X-ray spectrum because its energies are fixed and element-specific. Bremsstrahlung, on the other hand, forms a continuous spectrum, with intensity gradually increasing to a peak and then falling off. The combination of both gives the overall shape of the X-ray spectrum used in imaging.

Q2: Windowing (5 Points)

Below is a histogram showing the distribution of intensity values (in Hounsfield Units) from a CT image of a patient's chest. The x-axis represents the Hounsfield Units (HUs), and the y-axis represents the frequency of each HU value.



- Define a suitable window to best visualize the soft tissue (values around zero). Specify the minimum and maximum Hounsfield Units (HUs) for this window (2 Points)

minimum: -500 maximum: 500

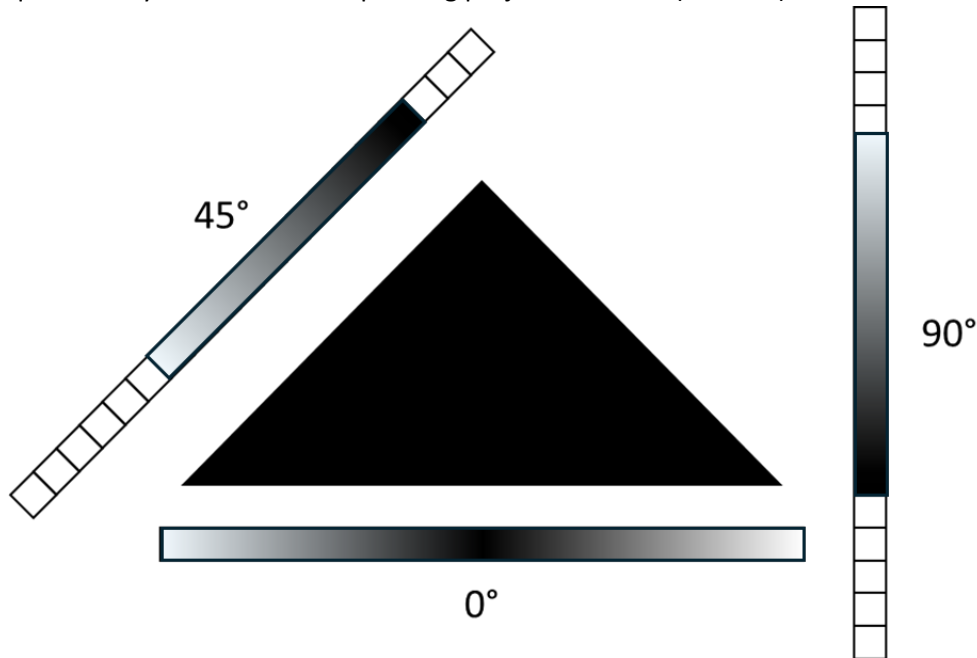
- Explain your reasoning for selecting this window. Discuss how the chosen window settings will help in enhancing the visualization of soft tissue and differentiate it from other structures (such as the lung) (3 Points):

The Hounsfield window in CT imaging is used to adjust the range of Hounsfield units (HU) displayed (1 point), enhancing contrast for specific tissues. Structures outside the selected HU range are displayed as pure black or white (1 point), reducing distraction from irrelevant tissues like bone or air (in the lung) (1 point).

Q3: CT (5 Points)

Below is the depiction of a triangle that will be scanned using a virtual CT scanner. The task is to sketch the projection of this triangle at three different angles (0° , 45° , and 90°).

- Sketch the projection of the triangle at the following angles: 0° , 45° , and 90° . For each angle, qualitatively shade the corresponding projection below (2 Points)



- What do you call a matrix that consists of these projections as columns? (1 Point):

Sinogram (1 point)

- Describe how the triangle could be reconstructed from these projections (2 Points):

Applying so called back projection.

Iteratively "smear back" projections (1 point) with the same angle θ (1 point) as during recording and add up.

At the end, normalize and filter the reconstructed image.

Q4: SPECT vs PET (3 Points)

What is the main difference between SPECT and PET?

The main difference lies in the type of radiation they detect:

SPECT detects single gamma photons (1 point), while PET detects pairs of photons (1 point) produced when positrons emitted by the tracer annihilate with electrons (1 point).